

Math Hard Question 12

1

★ Mark for Review

ABC

In the given equation, a and b are non-zero constants. Which of the following expressions gives a solution to the equation?

$$7ax^2 + (7b - 12a)x - 12b = 0$$

A. $-\frac{a}{b}$

B. $-\frac{b}{a}$

C. $\frac{a}{b}$

D. $\frac{b}{a}$

2

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ABC

In the given equation, a and b are constants. The equation has infinitely many solutions. What is the value of b ?

$$\frac{ax + b}{3} = \frac{6}{17}x - \frac{13}{12}$$

A. $-\frac{18}{17}$

B. $\frac{18}{17}$

C. $-\frac{13}{4}$

D. $\frac{13}{4}$

3

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ABC

In the xy -plane, a parabola has x -intercepts at $(-17, 0)$ and $(37, 0)$. If the equation of the parabola is written as:

$$y = ax^2 + bx + c,$$

where a , b , and c are constants, which of the following expressions gives c in terms of a ?

A. $c = -1080a$

B. $c = -629a$

C. $c = \frac{a}{34}$

D. $c = 51a$

4

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ABC

In triangle ABC , the measure of angle B is 90° and BD is an altitude of the triangle. The length of $AB = 15$, and the length of AC is 23 greater than the length of AB . What is the value of $\frac{BC}{BD}$?

A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$

5

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ABC

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Akeyla opened one of these accounts with an original account balance of \$700.00 and did not make any other deposits or withdrawals.

- 4 months after Akeyla opened the account, the balance had increased by 0.6% of the original balance.
- 6 months after opening, the balance had increased by an additional 0.1% of the balance at the end of the first 4 months.
- Every 2 months after the first 6 months, the balance increased by an additional 0.1% of the balance 2 months before.

Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 4$?

A. $B(x) = 704.20(1.001)^{\frac{x-4}{2}}$

B. $B(x) = 700.00(1.001)^{\frac{x-4}{2}}$

C. $B(x) = 704.20(1.001)^{2(x-4)}$

D. $B(x) = 700.00(1.001)^{2(x-4)}$

6

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ABC

The area of a rectangular region is increasing at a rate of 180 square centimeters per hour. What is this rate, in square meters per minute?

A. 3

B. 0.3

C. 0.03

D. 0.0003

7

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ABC

The exponential function $f(x)$ is defined by

$$f(x) = ab^x$$

where a and b are positive constants.

If

$$f(n-1) = f(n) + \frac{82}{100}f(n-1)$$

where n is a constant, what is the value of b ?

A. 1

B. 0.82

C. 0.18

D. 10

8

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ABC

The function f is defined by the given equation, where k is a constant. The value of $f(x)$ increases by 83% for every increase of x by 1.

$$f(x) = k(1.83)^x$$

For which of the following functions, where k is a constant, does the value of $g(z)$ increase by 83% for every increase of x by $\frac{1}{4}$?

A. $g(z) = k(1.83^x)^{\frac{1}{4}}$

B. $g(z) = k(1.83)^{x+\frac{1}{4}}$

C. $g(z) = k(1.83^x)^4$

D. $g(z) = k(1.83)^{x-4}$

9

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ABC

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the bases of each of the two pyramids is 100 cm^2 .

Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2 ?

A. 209**B.** 220**C.** 367**D.** 551